

MalGuard AI - Machine Learning Model Training Report

This report provides evaluation metrics and features summary for the trained classifiers.

1. Best Model Selection Summary

Based on model comparisons, the **Gradient Boosting** model was selected for production as it achieved the highest accuracy.

Model Name	Accuracy	Precision	Recall	F1 Score
Random Forest	89.1087%	0.8906	0.8911	0.8892
Extra Trees	81.5623%	0.8185	0.8156	0.8012
Gradient Boosting	91.5970%	0.9155	0.9160	0.9150

2. Explainable AI - Feature Importance Mapping

The relative importance score for each PE metadata feature extracted during training:

Feature Name	Importance Score
entropy	0.2237
has_suspicious_apis	0.2117
file_size	0.1662
num_imported_dlls	0.1393
num_imported_apis	0.1046
compile_timestamp	0.0716
num_sections	0.0673
has_suspicious_sections	0.0155

3. Class-specific Metrics (Gradient Boosting)

Malware Class	Precision	Recall	F1 Score
Benign	0.9809	0.9824	0.9817
Trojan	0.9686	0.9877	0.9780

Worm	0.8462	0.8691	0.8575
Spyware	0.7845	0.7728	0.7786
Ransomware	1.0000	1.0000	1.0000
Adware	0.8314	0.7167	0.7698
Backdoor	0.8196	0.9286	0.8707
Downloader	0.9951	0.9951	0.9951
Cryptominer	0.9637	0.9803	0.9719
Botnet	0.8286	0.7877	0.8076